

SALGIC (Salbutamol Injection BP 0.5 mg/ml, 1 ml Ampoule)

Description

Salbutamol Injection is a colorless or very pale yellow solution.
Each ml contains Salbutamol Sulfate BP Equivalent to Salbutamol 0.5mg.

Description after dilution

Salbutamol Injection 0.5 mg/ml on dilution with diluent 5% Dextrose in 0.9% Sodium Chloride Injection is clear colorless solution at 30°C and shelf life is 24 hours.

Pharmacodynamics

Salbutamol is a selective β_2 adrenoceptor agonist. At therapeutic doses it acts on the β_2 adrenoceptors in the bronchi and uterus, with little or no action on the β_1 adrenoceptors of the heart. It is suitable for the management of an asthmatic attack, and for uncomplicated premature labour, under the direction of a physician.

Pharmacokinetics

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally and partly by metabolism to the inactive 4'-O -sulphate (phenolic sulphate) which is also excreted primarily in the urine.
The faeces are a minor route of excretion. The majority of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%.

Indication

Relief of severe bronchospasm associated with asthma or bronchitis and for the treatment of status asthmaticus. Management of uncomplicated premature labour in the last trimester of pregnancy.

Contraindication

Salbutamol parenteral preparations are contraindicated in patients with a history of hypersensitivity to any of their components.
Although intravenous *Salbutamol* used in the management of premature labour, uncomplicated by conditions such as placenta praevia, ante-partum haemorrhage or toxemia of pregnancy, *Salbutamol* presentations should not be used for threatened abortion.

Side effects

Adverse reactions are listed below by system organ class and frequency.

Immune system disorders

Very rare: Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse.

Metabolism and nutrition disorders

Rare: Hypokalaemia
Potentially serious Hypokalaemia may result from beta 2 agonist therapy.
Unknown: Lactic acidosis
Lactic acidosis has been reported very rarely in patients receiving high therapeutic doses of intravenous and nebulised salbutamol therapy for the treatment of acute asthma exacerbation.

Nervous system disorders

Very common: Tremor
Common: Headache
Very rare: Hyperactivity

Cardiac disorders

Very common: Tachycardia.
Common: Palpitations
Rare: Cardiac arrhythmias including atrial fibrillation, Supraventricular tachycardia and extrasystoles.
Unknown: Myocardial Ischaemia.

Vascular disorders

Rare: Peripheral vasodilatation

Respiratory, thoracic and mediastinal disorders

Uncommon: Pulmonary oedema
In the management of pre-term labour, Salbutamol Injection /Solution for Intravenous Infusion have uncommonly been associated with pulmonary oedema; in some cases this has proved fatal. Patients with predisposing factors including multiple pregnancies, fluid overload, pre-existing cardiac disease, maternal infection and pre-eclampsia may have an increased risk of developing pulmonary oedema.

Gastrointestinal disorders

Unknown: Nausea, vomiting.
In the management of premature labour, intravenous infusion of Salbutamol Injection has very rarely been associated with nausea and vomiting.

Musculoskeletal and connective tissue disorders

Common: Muscle cramps.

Injury, poisoning and procedural complications

Unknown: Slight pain or stinging on intramuscular use of undiluted injection.

Warning and precautions

The management of asthma should normally follow a stepwise programme, and patient response should be monitored clinically and by lung function tests.

Increasing use of short acting inhaled β_2 agonists to control symptoms indicates deterioration of asthma control. Under these conditions, the patient's therapy plan should be reassessed. Sudden and progressive deterioration in asthma control is potentially life threatening and consideration should be given to starting or increasing corticosteroid therapy. In patients considered at risk, daily peak flow monitoring may be instituted.

The use of *Salbutamol* parenteral preparations in the treatment of severe bronchospasm or status asthmaticus does not obviate the requirement for glucocorticoid steroid therapy as appropriate.
When practicable, administration of oxygen concurrently with parenteral *Salbutamol* is recommended, particularly when it is given by intravenous infusion to hypoxic patients.

In common with other β -adrenoceptor agonists, Salbutamol can induce reversible metabolic changes such as reversible hypokalaemia and increased blood glucose levels. The diabetic patient may be unable to compensate for this and the development of ketoacidosis has been reported. Concurrent administration of corticosteroids can exaggerate this effect.

Potentially serious hypokalaemia may result from β_2 agonist therapy mainly from parenteral and nebulized administration. Particular caution is advised in acute severe asthma as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids, diuretics and by hypoxia. It is recommended that serum potassium levels are monitored in such situations.

Diabetic patients and those concurrent receiving corticosteroids should be monitored frequently during intravenous infusion of *Salbutamol* so that remedial steps (e.g. an increase in insulin dosage) can be taken to counter any metabolic change occurring. For these patients *Salbutamol Solution* for Intravenous Infusion should be diluted with Sodium Chloride Injection BP, rather than Sodium Chloride and Dextrose Injection BP.

Lactic acidosis has been reported very rarely in association with high therapeutic doses of intravenous and nebulised short-acting beta-agonist therapy, mainly in patients being treated for an acute asthma exacerbation (see Adverse Reaction section). Increase in lactate levels may lead to dyspnoea and compensatory hyperventilation, which could be misinterpreted as a sign of asthma treatment failure and lead to inappropriate intensification of short-acting beta-agonist treatment. It is therefore recommended that patients are monitored for the development of elevated serum lactate and consequent metabolic acidosis in this setting.

Salbutamol should be administered cautiously to patients with thyrotoxicosis.

Special Labelling Requirements as per DRDG:

Tocolysis: Serious adverse reactions including death have been reported of salbutamol to women in labour. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

As maternal pulmonary oedema and myocardial ischaemia have been reported during or following treatment of premature labour with β_2 agonists, careful attention should be given to fluid balance and cardio-respiratory function, including ECG should be monitored. If signs of pulmonary oedema or myocardial ischaemia develop, discontinuation of treatment should be considered.

Due to the risk of pulmonary oedema and myocardial ischaemia that has been observed, during the use of beta2-agonists

in the treatment of premature labour, before these products are given to any patient with known heart disease, an adequate assessment of the patients' cardiovascular status should be made by a physician experienced in cardiology.

Cautious use of salbutamol injections is required in pregnant patients when it is given for relief of bronchospasm so as to avoid interference with uterine contractility. During IV infusion of salbutamol, the maternal pulse rate should be monitored and not normally allowed to exceed a steady rate of 140 beats per minute.

Interactions
Salbutamol Injection and non-selective β -blocking drugs, such as propranolol, should not usually be prescribed together.

Pregnancy and Lactation
Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus.
During worldwide marketing experience, rare cases of various congenital anomalies, including cleft palate and limb defects have been reported in the offspring of patients being treated with salbutamol. Some of the mothers were taking multiple medications during their pregnancies.
Because no consistent pattern of defects can be discerned, and baseline rate for congenital anomalies is 2-3%, a relationship with salbutamol use cannot be established.
As salbutamol is probably secreted in breast milk its use in nursing mothers is not recommended unless the expected benefits outweigh any potential risk. It is not known whether salbutamol in breast milk has a harmful effect on the neonate.

Recommended dose
Method of administration
Salbutamol Injection may be administered by the subcutaneous (SC), intramuscular (I.M.) and intravenous (I.V.) injection and infusion.
Salbutamol has duration of action of 4 to 6 hours in most patients.
Increasing use of β_2 agonists may be a sign of worsening asthma. Under these conditions a reassessment of the patient's therapy plan may be required and concomitant glucocorticosteroid therapy should be considered.
Note: The contents of the ampoules of Salbutamol Solution for Intravenous Infusion must not be injected undiluted. The concentration should be reduced by 50% before administration.
Salbutamol parenteral preparations should not be administered in the same syringe or infusion as any other medication.

Adults
In severe bronchospasm and status asthmaticus
Subcutaneous Route:
500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.
Intramuscular Route:
500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.
Intravenous Route:
250 micrograms (4 micrograms /kg bodyweight) injected slowly. If necessary the dose may be repeated.
Infusion:
In status asthmaticus, infusion rates of 3 to 20 micrograms per minute are generally adequate but in patients with respiratory failure, higher dosage has been used with success. A starting dose of 5 micrograms per minute is recommended with appropriate adjustment in dosage according to patient response.
A suitable solution for infusion may be prepared by diluting with 5% Dextrose in 0.9% NaCl for Intravenous Infusion.
Dilution Instruction
A suitable solution for infusion may be prepared by diluting 10mL (5mg) of Salbutamol Solution for Intravenous Infusion in 500mL of an infusion solution such as sodium chloride and dextrose injection BP to provide a salbutamol dose of 10 μ g (0.01mg)/mL of solution.

In the management of premature labour
For this indication Salbutamol Solution for Intravenous Infusion is recommended using a solution prepared as above. Infusion rates of 10-45 micrograms per minute are generally adequate to control uterine contractions but greater or lesser infusion rates may be required according to the strength and frequency of contractions. A starting rate of 10 micrograms per minute is recommended, increasing the rate at 10-minute intervals until there is evidence of patient response shown by diminution in strength, frequency or duration of contractions.
Thereafter the infusion rate may be increased slowly until contractions cease. Careful attention should be given to cardio-respiratory function and fluid balance monitoring. The maternal pulse rate should be monitored and the infusion rate adjusted to avoid excessive maternal heart rates (above 140 beats per minute). Treatment discontinuation should be considered should signs of pulmonary oedema or myocardial ischaemia develop (see Warnings and Precautions and Adverse Reactions).
Once uterine contractions have ceased the infusion rate should be maintained at the same level for one hour and then reduced by 50% decrements at 6 hourly intervals. Treatment may be continued orally with Salbutamol Tablets 4 milligram given three or four times daily.
As an alternative procedure or to counteract inadvertent over dosage with oxytocic drugs, Salbutamol Injection may be administered as a single injection by the intravenous or intramuscular routes. The usual recommended dose is 100 to 250 micrograms of salbutamol. The dose may be repeated according to the response of the patient.

Children
At present there is insufficient evidence to recommend a dosage regimen for routine use in children.

Over dosage
The most common signs and symptoms of overdose with salbutamol are transient beta agonist pharmacologically mediated events (see Warnings and Precautions and Adverse Reactions).
Hypokalaemia may occur following overdose with Salbutamol. Serum potassium levels should be monitored. Nausea, vomiting and hyperglycaemia have been reported, predominantly in children and when salbutamol overdose has been taken via the oral route.
Treatment of over dosage:
Consideration should be given to discontinuation of treatment and appropriate symptomatic therapy such as cardio-selective beta-blocking agents in patients presenting with cardiac symptoms (e.g. tachycardia, palpitations). Beta-blocking drugs should be used with caution in patients with a history of bronchospasm.

Incompatibilities
None.

Storage condition
Store at a temperature below 30°C and protect from light.

Storage condition after dilution
Salbutamol Injection 0.5 mg/ml on dilution with diluent 5% Dextrose in 0.9% Sodium Chloride Injection is clear colorless solution at 30°C and shelf life is 24 hours.

Product Shelf life
36 months

After dilution Shelf life
24 hours at 30°C

Presentation
Packs of 1mL x 10 Ampoules.

Manufactured by:
Gland Pharma Limited
Unit-II, Plot No. 42 to 52,
Sy.No.166, 171, 172 & 177,
Phase – III, TSIC Pashamylaram (V),
Patancheru (M),
Sangareddy District - 502 307,
Telangana, INDIA.

Product Registration Holder:
UNIMED SDN BHD
No. 53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara,
52200 Kuala Lumpur, Malaysia
Revised date: 26 April 2019

LEA-xxxxxx-00